# NAM KAIO HYDRO POWER PROJECT

| | |
|---|---|
| Project title | NAM KAIO HYDRO POWER PROJECT |
| Project ID | / |
| Crediting period | 31-Mar-2026 to 30-Mar-2036 (expected crediting period) |
| Original date of issue | 20-Dec-2025 |
| Most recent date of issue | 20-Dec-2025 |
| Version | 1.0 |
| *VCS Standard* Version | 4.7 |
| Prepared by | QIAN LIU, SINGAPORE TREASURE CARBON TECH PTE. LTD. |

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# 1 PROJECT DETAILS

## 1.1 Summary Description of the Project

Nam Kaio hydro power project (hereafter referred to as the “the project”) is developed by Nam Kiao Hydropower Sole Co., Ltd. in Muong Mok District, Xieng Khouang Province, Lao PDR. The primary purpose of the project is to generate electricity to national grid of Vietnam.

The project is an impoundment hydropower with the total installed capacity of 11 MW, provided by two Pelton turbines with horizontal axes and the unit capacity is 5.5 MW. The project is expected to be commissioned on 31/03/2026 with an average annual output of 36,530 MWh.

The project is expected to constantly contribute clean energy to the Vietnam Power Grid. It not only conforms to the present situation, but also can adapt to future changes in the electricity market.

The baseline scenario of the project is continuation of the present situation, i.e electricity supplied from the power grid. By displacing part of the power generated by thermal power plants, the project is therefore expected to reduce $CO_2$ emissions by an estimation of 24,080 $tCO_2e$ per year during the fixed 10 years crediting period.

The project is not only supplying renewable electricity to grid, but also producing positive environmental and economic benefits and also promotes the local sustainable development in the following aspects:

• During the construction and operation of the project, it would bring a large number of job opportunities to the local people, especially those from poor families. Then it will help them earn income through working in the project, thus contributing to the economic stability of the local community.

• The roads, bridges, and communication facilities built for the construction of the hydro power plant have significantly improved transportation and communication conditions in the project area, facilitating easier travel for residents and more efficient transport of goods.

• The project provides a stable and reliable electricity supply to meet Vietnam's rapidly growing industrial and residential demand, thereby alleviating the pressure of power shortages.

• By replacing electricity generated from fossil fuels on the Vietnamese grid with renewable power, the project reduces GHG emissions.

## 1.2 Audit History

| Audit type | Period | Program | Validation/verification body name | Number of years |
|---|---|---|---|---|
| *Validation/ verification* | To Be Determined | VCS | To Be Determined | To Be Determined |

## 1.3 Sectoral Scope and Project Type

| | |
|---|---|
| Sectoral scope[1] | 01- Energy Industries (Renewable/non-renewable sources) |
| Project activity type | Renewable Energy Project |

## 1.4 Project Eligibility

### 1.4.1 General eligibility

According to section 3.1 of the VCS Standard version 4.7, establishing a consistent and standardized certification process is critical to ensuring the integrity of VCS projects. Accordingly, certain high-level requirements must be met by all projects, as set out below.

| Reference to VCS Standard version 4.7 | Requirement | Justification |
|---|---|---|
| 3.1.1 | Projects shall meet all applicable rules and requirements set out under the VCS Program, including this document. Projects shall be guided by the principles set out in Section 2.2.1. | The project meets all applicable rules and requirements as set out under the VCS Program. |
| 3.1.2 | Projects shall apply methodologies eligible under the VCS Program. Methodologies shall be applied in full, including the full application of any tools or modules referred to by a methodology, noting the exception set out in Section 3.14.1. The list of methodologies and their validity periods is available on the Verra website. | The project applies CDM approved methodology AMS-I.D. version 18.0 which is an eligible VCS methodology along with tool or modules as applicable. Refer to section 3.1 and 3.2 of the PD below for more details. |

[1] *Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes*

| | | |
|---|---|---|
| 3.1.3 | Projects shall apply the latest version of the applicable methodology in all cases unless a grace period applies to the project as set out in 3.22 below. Projects must update to the latest version of the methodology when reassessing the baseline and renewing a crediting period. | The project applies latest version of AMS-I.D. (version 18.0), and type of ten years fixed period is selected. It does not need to renew a fixed crediting period. |
| 3.1.4 | Projects and the implementation of project activities shall not lead to the violation of any applicable law, regardless of whether or not the law is enforced. | Projects are in compliance with currently applicable laws. According to the approval from the Department of Energy Policy and Planning of the Ministry of Energy and Mines, the project complies with all relevant laws and regulations. Refer to section 1.15 of the PD below for more details. |
| 3.1.5 | Where projects apply methodologies that permit the project proponent its own choice of model (see the VCS Program document VCS Program Definitions for the definition of model), the model shall meet the requirements set out in the VCS Methodology Requirements, and it shall be demonstrated at validation that the model is appropriate to the Project circumstances (i.e., use of the model will lead to an appropriate quantification of GHG emission reductions or carbon dioxide removals). | Not applicable. There are no model needs to be chosen by the project proponent as per the applied methodology AMS-I.D. version 18.0. |
| 3.1.6 | Where projects apply methodologies that permit the project proponent to choose a third-party default factor or standard to | The default factor or standard applied in this project are all methodologically permissible, so the default factor or standard |

| | | |
|---|---|---|
| | ascertain GHG emission data and any supporting data for establishing baseline scenarios and demonstrating additionality, such default factor or standard shall meet the requirements set out in the VCS Methodology Requirements. | meet the requirements set out in the VCS Program document VCS Methodology Requirements. More details see Section 5.1. |
| 3.1.7 | Where the rules and requirements under an approved GHG program conflict with the rules and requirements of the VCS Program, the rules and requirements of the VCS Program shall take precedence. | GWP values used in this project are based on the IPCC AR5 revised in 2019, meets the rules and requirements of the VCS standard. The default parameters used in this project based on 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories. |
| 3.1.8 | Where projects apply methodologies from approved GHG programs, they shall conform with any specified capacity limits (see the VCS Program Definitions for the definition of capacity limit) and any other relevant requirements set out with respect to the application of the methodology and/or tools referenced by the methodology under those programs. | Small-scale methodology AMS-I.D. version 18.0 is applied by the project, which is approved by CDM programs, and the annual emission reductions of the project is less than 60 kt CO2 which satisfies the requirement on capacity limit of the small-scale methodology AMS-I.D. version 18.0.<br><br>And then the application of the methodology and/or tools referenced by the methodology under the project activity is also applicable, please refer to section 3 for more details. So, it is applicable. |
| 3.1.9 | Where Verra issues new VCS Program rules, the effective dates of these requirements are set out in Appendix 3 Document History and | The fixed 10 years crediting period was chosen for the project, and it has followed all the latest requirements of Verra. |

| | | |
|---|---|---|
| | Effective Dates or equivalent for other program documents, and are listed in a companion Summary of Effective Dates document which corresponds with each update. | |

The project activity is 11 MW small scale hydro power project located in Laos PDR, a least developed country as per United Nation list of LDC. As per para 3.10.1 of VCS Standard V4.7, the project activity is under the category of "project" as it has estimated annual emission reduction potential less than 300,000 tonnes of $CO_2e$ per year. The project activity is eligible to use small scale approved methodology AMS-I.D. version 18.0. Therefore, in line with section 2.1 of VCS Standard version-4.7 the project activity is eligible to use VCS programme of VERRA.

As per para 3.8.1 of VCS Standard version 4.7, Non-AFOLU projects shall complete validation within two years of the project start date. The expected project start date is in March, 2026 and the project activity will be listed with VERRA for public comments on December 20, 2025, which is earlier than the project start date, complying with the VCS listing/validation completion requirements.

As per para 9 of TOOL 20, Assessment of Debundling for Small-Scale Project Activities" V4.1, A proposed small-scale project activity shall be deemed to be a debundled component of a large project activity if there is a registered small-scale CDM project activity or an application to register another small-scale CDM project activity:

a) With the same project participants;

b) In the same project category and technology/measure; and

c) Registered within the previous 2 years; and

d) Whose project boundary is within 1 km of the project boundary of the proposed small- scale

activity at the closest point.

The proposed project is yet to be commissioned and Nam Kiao Hydropower Sole Co., Ltd. has no other small-scale or large-scale project commissioned within 1 km of the project boundary, hence the proposed project activity is not a fragmented part of the larger project activity.

| VCS Scope | | Justification |
|---|---|---|
| 2.1.1 | The scope of the VCS Program includes:<br>a) The seven Kyoto Protocol greenhouse gases.<br>b) Ozone-depleting substances (ODS).<br>c) Project activities supported by a methodology approved under the VCS | a) The project activity aims to reduce $CO_2e$ emissions, which are GHG gases identified under Kyoto Protocol<br>b) Not applicable<br>c) Not applicable |

| | | |
|---|---|---|
| | Program through the methodology development and review process.<br>d) Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions).<br>e) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements. | d) The project activity uses CDM approved small scale methodology AMS-I.D., the CDM GHG programme is approved by VCS. Hence criterion is met.<br>e) Not applicable |
| 2.1.2 | The scope of the VCS Program excludes projects that can reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal, or destruction. | The project activity has not been implemented to generate GHG emissions for the purpose of their subsequent reduction, removal or destruction. Hence criterion is met. |
| 2.1.3 | The VCS Program also excludes the project activities under the circumstances indicated in Table 1 of clause 2.1.3 of the VCS Standard version-4.7 | The project activity is 11 MW small scale hydro power project located in Laos PDR a least developed country [2] as per United Nation list of LDC. In line with guidance under table 1, the project activity is not under exclusion category. |

### 1.4.2 AFOLU project eligibility

N/A.

### 1.4.3 Transfer project eligibility

N/A.

## 1.5 Project Design

☒ Single location or installation

[2] https://www.un.org/en/conferences/least-developed-countries#:~:text=The%2046%20countries%20currently%20on,%2C%20Kiribati%2C%20Lao%20People's%20Dem.

☐ Multiple locations or project activity instances (but not a grouped project)

☐ Grouped project

### 1.5.1 Grouped project design

The project activity is not a grouped project. Thus, this section is not applicable.

## 1.6 Project Proponent

| | |
|---|---|
| Organization name | SINGAPORE TREASURE CARBON TECH PTE. LTD. |
| Contact person | Qian Liu |
| Title | Project manager |
| Address | 24 SIN MING LANE, #06-97, MIDVIEW CITY, SINGAPORE 573970 |
| Telephone | +86-18758409675 |
| Email | liuqian@treasurecarbon.com |

## 1.7 Other Entities Involved in the Project

This section will be completed in the full Project Description(PD) submitted for the 'under validation' status, as this version of the PD is for pipeline listing purposes only.

| | |
|---|---|
| Organization name | |
| Role in the project | |
| Contact person | |
| Title | |
| Address | |
| Telephone | |
| Email | |

## 1.8 Ownership

The project owner is Nam Kiao Hydropower Sole Co., Ltd., who has the legal right to control and operate the project activities. The business licence of the project owner are the evidences for legislative right. Besides, the project allotment license, the equipment purchasing contract and the power purchase agreement are the evidence for the ownership of the plant equipment and power generating.

The project owner authorizes Singapore Treasure Carbon Tech Pte. Ltd. as the project proponent to develop and manage the VCS project and account .

## 1.9 Project Start Date

| | |
|---|---|
| Project start date | 31-Mar-2026 |
| Justification | As per 3.8 of VCS Standard (Version 4.7), the project start date of a non-AFOLU project is the date on which the project began generating GHG emission reductions or carbon dioxide removals. The project activity is a non-AFOLU project. The construction of the project is expected to be completed in 31-Mar-2026 and it is expected to begin commercial operation on 31- Mar-2026.<br><br>Thus, the project start date is 31-Mar-2026. |

## 1.10 Project Crediting Period

| | |
|---|---|
| Crediting period | ☐ Seven years, twice renewable<br><br>☒ Ten years, fixed<br><br>☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements) |
| Start and end date of first or fixed crediting period | 31-Mar-2026 to 30-Mar-2036 (inclusive of both start and end date) |

## 1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☒ < 300,000 tCO2e/year (project)

☐ ≥ 300,000 tCO2e/year (large project)

| Calendar year of crediting period | Estimated GHG emission reductions or removals ($tCO_2e$) |
|---|---|
| 31-March-2026 to 31-December-2026 | 18,208 |

| | |
|---|---|
| 01-January-2027 to 31-December-2027 | 24,080 |
| 01-January-2028 to 31-December-2028 | 24,080 |
| 01-January-2029 to 31-December-2029 | 24,080 |
| 01-January-2030 to 31-December-2030 | 24,080 |
| 01-January-2031 to 31-December-2031 | 24,080 |
| 01-January-2032 to 31-December-2032 | 24,080 |
| 01-January-2033 to 31-December-2033 | 24,080 |
| 01-January-2034 to 31-December-2034 | 24,080 |
| 01-January-2035 to 31-December-2035 | 24,080 |
| 01-January-2036 to 30-March-2036 | 5,871 |
| **Total estimated ERRs during the first or fixed crediting period** | 240,799 |
| **Total number of years** | 10 |
| **Average annual ERRs** | 24,080 |

## 1.12 Description of the Project Activity

The project activity is a run-of-river Hydro power project with a total installed capacity of 11 MW provided by two Pelton turbines with horizontal axes.

The Project activity is a Greenfield project and the electricity generated by the project will be transmitted to the power grid of Lao PDR firstly and finally exported to the the power grid of Vietnam. The project will therefore displace an equivalent amount of electricity which would have otherwise been generated by fossil fuel dominant electricity grid.

In the Pre- project scenario, the entire electricity delivered to the grid by the project activity, would have otherwise been generated by the operation of grid-connected power plant. The scenario prior to the start of implementation of the project activity is generation of electricity, dominated by thermal power plants, thus leading to GHG emissions. The baseline scenario is the same as the scenario prior to the start of implementation of the project activity.

The project shall result in replacing anthropogenic emissions of greenhouse gases (GHG's) estimated to be approximately 24,080 $tCO_2e$ per year, thereon displacing 36,530 MWh/year amount of electricity from the gird over the 10 years crediting period.

The construction of the hydropower project involves diversion plan, dam/weir and appurtenant structure, water conveyance system, powerhouse, electro-mechanical equipments and transmis line. And the main equipment relevant to the electricity generation is listed on the below table.

**Table.1: Technical Parameters of the main project equipment**

| Turbine | | |
|---|---|---|
| **Item** | **Value** | **Unit** |
| Type of turbine | Pentol with Horizontal Shaft | / |
| Number of turbines | 2 | sets |
| Design discharge | 2 x 0.99 | $m^3/s$ |
| Design head | 581.33 | m |
| Runner diameter | 1.1 | m |
| Rate speed | 750 | r/min |
| Capacity of one unit | 5.5 | MW |
| **Generator** | | |
| **Item** | **Value** | **Unit** |
| Number of generators | 2 | sets |
| Installed Capacity | 2 x 5.5 | MW |
| Rated voltage | 6.3 | kV |
| Frequency | 50 | HZ |
| Rate speed | 750 | r/min |

## 1.13 Project Location

The project is construction on Nam Kiao stream, Mok District, Xieng Khouang Province, Lao PDR.

**Table.2: Project Location**

| Project Location | Site | Latitude | Longitude |
|---|---|---|---|
| Muong Mok District, Xieng Khouang Province, Lao PDR. | Dam | 19°10'43.5" | 103°51'56.5" |
| | Powerhouse | 19°11'12.6" | 103°53'12.3" |

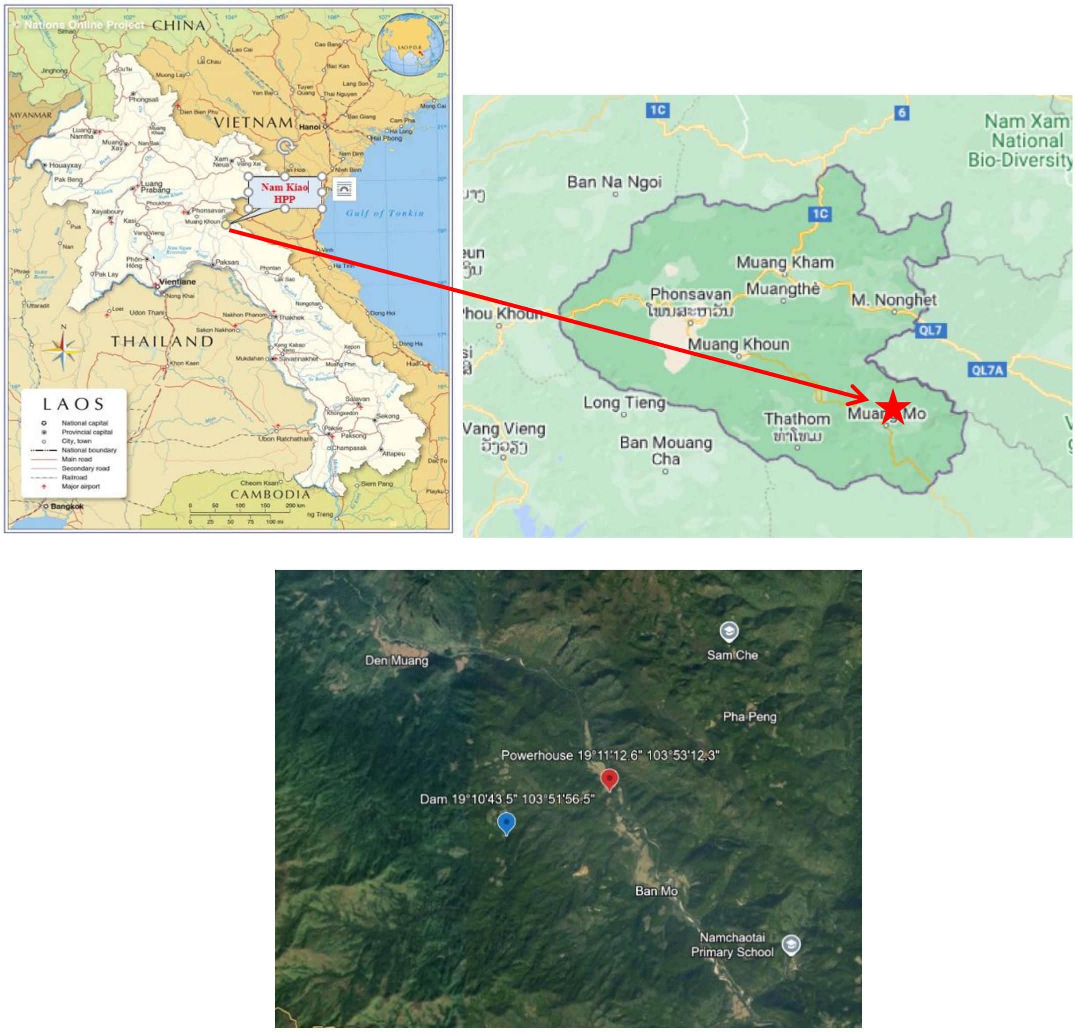


Figure.1: Project Site

## 1.14 Conditions Prior to Project Initiation

The project is a Greenfield hydro power project and does not involve generation of GHG emissions for the purpose of their subsequent reduction, removal or destruction. Thus prior to project initiation, there was nothing at site. The project activity supply electricity to national grid of Vietnam and in absence of proposed project the equivalent amount of electricity would have continued to be generated in grid mix power plants, which is also the baseline scenario. For more information refer to section 3.4 for the baseline scenario.

## 1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project has received all the necessary approvals for the development and commissioning of the proposed 11 MW hydro power project from the respective nodal agencies and is in compliance to the local laws and regulations. The list of applicable laws & regulations and the project compliance against it are given below:

<table>
<tr><th>Laws & Regulation</th><th>Project Compliance</th></tr>
<tr><td>Law on Electricity (amended version) No. 03/NA, dated 20/12/2011[3] ;</td><td rowspan="2">Project received the approval from the Department of Energy Policy and Planning of the Ministry of Energy and Mines that confirms the project compliance with these Laws & Regulation. This approval confirms the project's compliance with these laws and regulations.<br><br>The project also signed PPA with Vietnam Electricity, a state-owned enterprise which is also an approval for electricity connection with the grid.</td></tr>
<tr><td>Decree on the Approval and Promulgation of the Policy on Sustainable Hydropower Development of the Lao PDR No. 02/GoL., dated 12/01/2015[4] ;</td></tr>
<tr><td>Environmental Protection Law (revised version), No. 29/NA, dated 18/12/2012[5];</td><td rowspan="2">Project received the Environmental Certificate from the Provincial Department of Natural Resources and Environment, based on the submitted Initial Environmental Examination (IEE) Report and the Environmental and Social Management and Monitoring Plan (ESMMP). This certificate confirms the project's compliance with these laws and regulations</td></tr>
<tr><td>National Policy on Environmental and Social Sustainability of the Hydropower Sector in Lao PDR[6].</td></tr>
</table>

## 1.16 Double Counting and Participation under Other GHG Programs

### 1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

[3] https://policy.asiapacificenergy.org/sites/default/files/Electricity%20Law.pdf

[4] policy.asiapacificenergy.org/sites/default/files/Decree%20on%20the%20Approval%20and%20Promulgation%20of%20the%20Policy%20on%20Sustainable%20Hydropower%20Development%20in%20Lao%20PDR.pdf

[5] https://www.ajcsd.org/chrip_search/dt/pdf/Useful_materials/Lao_PDR/Environmental_Protection_Law_Revised_Version.pdf

[6] policy.asiapacificenergy.org/sites/default/files/National%20Policy%20on%20the%20Environmental%20and%20Social%20Sustainability%20of%20the%20Hydropower%20Sector.pdf

☐ Yes ☒ No

### 1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

### 1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

## 1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

### 1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

### 1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

### 1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

*If yes:*

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

*If yes:*

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes ☒ No

## 1.18 Sustainable Development Contributions

The project activity implemented by the project owner can contribute to sustainable development as defined by and tracked against the United Nations Sustainable Development Goals (SDGs). The specific analysis is as follows:

**SDG 7: Affordable and Clean Energy**

The project contributes SDG Target 7.2.1 "By 2030, increase substantially the share of renewable energy in the global energy mix" by the utilization of hydropower as a renewable energy source. By generating an estimated annual output of 36,530 MWh of renewable energy, this contribution aligns with the overarching goal of fostering a greener and more sustainable future.

Thus, the project will contribute to SDG 7 Affordable and Clean Energy.

**SDG 8: Decent Work and Economic Growth**

The project creates direct and indirect employment opportunities during construction and operation phases, so it contributes to SDG Target 8.5.2 "By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities and equal pay for work of equal value". The project activity is expected to create some regular employment during operation.

Thus, the project will contribute to SDG 8 Decent Work and Economic Growth.

**SDG 13: Climate Action**

The project produces clean renewable energy by diminishing $CO_2$ emissions. Therefore, it contributes SDG Target 13.2.2 "Total greenhouse gas emission per year". The project activity expected to reduce 24,080 $tCO_2e$ per year.

Thus, the project will contribute to SDG 13 Climate Action.

## 1.19 Additional Information Relevant to the Project

### 1.19.1 Leakage Management

Not applicable as no leakage emission is involved in this project activity.

### 1.19.2 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

### 1.19.3 Further Information

Not applicable, as this project activity does not involve any additional information that may have a bearing on the eligibility of the project, the GHG emission reductions, or the quantification of the project’s reductions.

# 2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

## 2.1 Stakeholder Engagement and Consultation

### 2.1.1 Stakeholder Identification

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|---|---|
| Stakeholder Identification | Stakeholders include various individuals, groups, and institutions who have an interest in, could be impacted by the project, possess the ability to significantly influence, or are crucial for achieving the desired project outcome. The list of relevant stakeholders identified through stakeholder mapping includes:<br>• The investor of the project<br>• Employees<br>• Local villagers<br>• Local village-level mass organizations<br>• Local policymakers and representatives of local authorities.<br>Stakeholders have been invited through various means of invitation like public announcement, invitation letters, etc. |
| Legal or customary tenure/access rights | The project is built own and legally owned by Nam Kiao Hydropower Sole Co., Ltd. The dam is built on the crop rotation farmland and the powerhouse is within the forest reserve in the village. Thus, the land |

| | |
|---|---|
| | within the project boundary was legally classified as state-owned land. While there was no private ownership of the land by local residents, communities in the vicinity historically practiced customary access to specific sections of the project area. These areas were utilized for the collection of wild edibles and natural resources, such as fruit, mushrooms, herbal medicines and nuts etc., which supported local subsistence.<br><br>Acknowledging the economic value of these natural resources to the local households, the project owner has provided fair compensation for the loss of access and potential income, in full compliance with national regulations and the project's resettlement and compensation plan.<br><br>Following the completion of the compensation process, all customary claims have been effectively settled. The project owner currently holds 100% clear legal and customary rights over the project site during the entire crediting period, with no ongoing disputes regarding land tenure or access. |
| Stakeholder diversity and changes over time | The Nam Kiao Hydro power project is located in Muang Mok District, Xieng Khouang Province. A socio-economic survey was carried out at Muang Mok District, Xieng Khouang Province to determine relevant demographic characteristics of the communities. Based on the socio-economic profile, stakeholder mapping was carried out to ensure that equitable, nondiscriminatory, and meaningful consultation was carried out.<br><br>The stakeholder group exhibited notable diversity across social, economic and cultural dimensions. Socially, individuals from various backgrounds, including different age groups, ethnicities and socioeconomic classes were present. Overtime, the composition of these groups may undergo changes due to evolving circumstances and development.<br><br>However, project activity does not lead to any stakeholder’s diversity as the hydro power project does not have any direct or indirect impact on the local stakeholder culture or diversity.<br><br>To ensure that the project continues to align with the diverse needs and expectations of its stakeholders, the project proponents has implemented a Grievance Redress Mechanism (GRM). |

| | |
|---|---|
| Expected changes in well-being | The implementation of the project is expected to have a positive impact on the well-being and other characteristics of stakeholders. Relative to the baseline scenario, the project will bring about the following changes:<br><br>**Climate change Impact**<br><br>During the entire construction and operation process of the project, almost no greenhouse gas emissions were produced, the project activity supplies net electricity generated to grid, thereby results in emission reduction due to displacing equivalent electricity from grid dominated by fossil fuel based thermal power plants, thereby reducing the impact of the project on the climate's greenhouse gases. Moreover, the wastewater, noises and solid waste generated by the project were all properly handled, meeting national standards.<br><br>**Social welfare**<br><br>The implementation of the project will promote local economic development, provide employment opportunities, and promote the prosperity of the rural economy.<br><br>Collectively, these changes will have a positive impact on residents' livelihoods and living environment, while enhance the sustainability of local communities. |
| Location of stakeholders | The location of the stakeholders is Muang Mok District, Xieng Khouang Province. The project is run of the river project; hence, project has no impact on stakeholders' field nearby or by other means. |
| Location of resources | The location of territories and resources is located in Muang Mok District, Xieng Khouang Province.<br><br>Other than project owners and working employees no other general stakeholders or local communities have access to the project site. And adequate food is supplied to workers to prevent local natural resources from being used as food. |

### 2.1.2 Stakeholder Consultation and Ongoing Communication

<table>
<tr><td>Date of stakeholder consultation</td><td>23-March-2021</td></tr>
<tr><td>Stakeholder engagement process</td><td>
Nam Kiao Hydropower Sole Co., Ltd. has identified relevant stakeholders as the investor of the project, employees, local villagers, local village-level mass organizations and local policymakers and representatives of local authorities. The invitation letters were sent to identified stakeholders well in advance dated 16-March-2021 to attend the meeting. The local villagers and the office bearers were also invited personally by a representative of the project proponent. The comments were invited in an open and transparent manner in the physical meeting by facilitating with a presentation on the proposed project activity.
During the meeting, the project proponent explained the benefits, possible costs and potential risks of the project to the participants. In addition, a survey of the local population was conducted. The survey was conducted through distributing and collecting responses to a questionnaire. The questionnaire investigated the impact of the project on the local area from the aspects of environment, economy and whether the surveyed people supported the project construction. About 50 people participate, gender sensitivity was ensured by the equal participation of men and women in the questionnaire. In total 50 out of 50 questionnaires were returned with a 100% response rate. Of the 50 people who participated in the survey, 27 were men and 23 were women, detailed information can be found in the table below:
<table>
<tr><th>No.</th><th colspan="2">Category</th><th>Amount</th></tr>
<tr><td rowspan="2">1.</td><td rowspan="2">Gender</td><td>Male</td><td>27</td></tr>
<tr><td>Female</td><td>23</td></tr>
<tr><td rowspan="3">2.</td><td rowspan="3">Age</td><td>⩽25</td><td>2</td></tr>
<tr><td>26~54</td><td>40</td></tr>
<tr><td>⩾55</td><td>8</td></tr>
<tr><td rowspan="2">3.</td><td rowspan="2">Educational background</td><td>⩽Junior High School</td><td>11</td></tr>
<tr><td>High school and junior college</td><td>29</td></tr>
</table>
</td></tr>
</table>

| | | | |
|---|---|---|---|
| | | Bachelor | 8 |
| | | Master | 2 |
| 4. | Identity | Resident of the project area | 20 |
| | | Local community representative / Village leader | 7 |
| | | Staff of local government department | 3 |
| | | Employee of the project investor/developer | 15 |
| | | Other (Please specify) | 5 |

**Consultation outcome**

In addition, FPIC (Free, prior, and informed consent) was discussed to ensure the legitimacy and transparency of the project. The project also discusses the VCS validation process to ensure the sustainability and environmental benefits of the project. Based on the results of the discussion at the meeting and the 50 questionnaires received for the survey, opinions are summarized as follows:

| No. | Question | Answer | Amount |
|---|---|---|---|
| 1 | Prior to this survey, how familiar were you with this hydropower project? | Very well known | 45 |
| | | Relatively understand | 5 |
| | | Don’t know | 0 |
| 2 | Do you think that the construction of the project will be beneficial to the local environment? | Yes | 50 |
| | | No | 0 |
| | | Don’t know | 0 |
| 3 | | Improve local infrastructure | 50 |

| | | | | |
|---|---|---|---|---|
| | | How do you think the construction of this project will affect the region? | Provide employment opportunities | 50 |
| | | | Destroy the surrounding environment | 10 |
| | 4 | Do you think the construction of this project will be beneficial to the economic development of the region? | Yes | 50 |
| | | | No | 0 |
| | | | Don't know | 0 |
| | 5 | Do you think that the construction of this project will have any adverse effects on your life, studies or work? | Yes | 0 |
| | | | No | 45 |
| | | | Don't know | 5 |
| | 6 | Do you support the construction of this project? | Yes | 47 |
| | | | No | 0 |
| | | | Not care | 3 |

The overall response to the project from all invited stakeholders, was encouraging and positive to local communities, seen as improving social-economic conditions. For the stakeholders who have some doubts of the proposed project or are lack of knowledge related to VCS or emission reduction project development, Nam Kiao Hydropower Sole Co., Ltd. made further explanations in response to their questions and dispel their concerns. In summary, there were no adverse reactions, comments or clarifications received during the stakeholder consultation process. There were also no comments or issues raised on the needs to make any changes to the project design.

| | |
|---|---|
| Ongoing communication | The project proponent has a grievance mechanism. Stakeholders can anytime provide their inputs or concerns over the project through any of the following means: |

| | |
|---|---|
| | • Complaint register: Stakeholders can provide the grievance in writing in the grievance register maintained at the site.<br><br>• Telephone access: Stakeholders can provide their grievances through calling site telephone.<br><br>• Email access: Stakeholders can provide their grievance through email using the email ID.<br><br>The site manager is responsible for addressing the grievances received from stakeholders. The same is informed during the stakeholder consultation meeting. This mechanism will ensure to continue to engage with local stakeholder in an ongoing proactive manner. |
| Stakeholder input | During the continuous communication with stakeholders, the project has not received any complaints, comments or suggestions from stakeholders. |

### 2.1.3 Free Prior and Informed Consent

| | |
|---|---|
| Obtaining consent | The project activity has been implemented with all necessary approvals from the Department of Energy Policy and Planning of the Ministry of Energy and Mines. Project proponent has conducted LSC during which all the necessary information regarding the project activity was explained to the stakeholders.<br><br>The project proponents fostered open dialogues, providing stakeholders ample opportunities to express concerns, ask questions, and share input. All raised concerns and inputs have been documented, and a grievance resolution mechanism is in place to address any ongoing conflicts, demonstrating the project's commitment to prioritizing resolution and community harmony. Regular stakeholder engagement will persist throughout the project lifecycle to address evolving concerns and ensure ongoing support.<br><br>Through this comprehensive process, the project not only obtained consent but also built a foundation of trust and cooperation with stakeholders, resulting in no ongoing or unresolved conflicts. |
| Outcome of FPIC | The result of the FPIC and stakeholder consultation is positive, with stakeholders expressing their support for the project. Project proponent has ensured that the project did not encroach on land, |

| | |
|---|---|
| | relocate people without consent, or cause forced physical or economic displacement. |

### 2.1.4 Grievance Redress Procedure

| | |
|---|---|
| **Development process** | Grievance Redress Mechanism (GRM) has been established to address concerns raised by stakeholders. This involves a Grievance Redress Committee (GRC), consisting of relevant commune and/or village chiefs, provincial officers or their nominees, a District Officer from the Cadastral office or their nominee, the Contractor, and a witness of the affected person, with at least one female member.<br><br>The GRC is tasked with (i) resolving land acquisition issues (if any), (ii) addressing concerns related to dust, noise, vibration, and construction-related nuisances to the public, (iii) convening monthly to review lodged complaints (if any), (iv) recording grievances and resolving them within 30 days from filing, and (v) reporting the status of grievance resolution and decisions/actions taken to the complainant(s)/affected persons.<br><br>The Provincial GRC, established at the provincial level, includes representation from commune councils across the project area of influence where project components will be implemented. Key contacts for the GRC have been prominently displayed at project office sites and public notice boards in affected communes. |
| **Grievance redress procedure** | The project participant has established a record keeping system and if any feedback/complain received shall be recorded and remedy to same will be done with information to concern person/entity. If any concern/feedback received will be discussed and if require stakeholder will be invited again to discuss and take necessary action as may be needed. |

### 2.1.5 Public Comments

| Comments received | Actions taken |
|---|---|
| To be added | To be added |

## 2.2 Risks to Stakeholders and the Environment

### 2.2.1 Management Experience

The project owner has long term experience in operating hydro power projects and has a proven track record for operation of similar projects in Host Country.

Our management team has the experience and expertise to effectively implement similar project activities. Our team members have many years of working experience in the field of hydro power and environmental protection, and expertise in project planning, implementation and supervision to effectively manage and drive the implementation of the project.

In addition, we have developed a recruitment strategy to fill gaps in skills and experience that may exist in the team. We will be looking for people with relevant experience and expertise to join the team to ensure that the project is supported and guided.

### 2.2.2 Risk Assessment

| | Risks identified | Mitigation or preventative measure(s) taken |
|---|---|---|
| Natural and human-induced risks to stakeholders' wellbeing | No risk identified | The implementation of the project and its impact on the natural environment comply with relevant laws and regulations and are expected to have a positive impact on the well-being of stakeholders. Therefore, the project has no natural and human induced risks to stakeholders' wellbeing. |
| Risks to stakeholder participation | No risk identified | No risks to stakeholder participation were identified during the preparation for stakeholder consultation. The project developer analyzed relevant stakeholders, risks, opportunities, and strategies for conducting the most effective consultation. To ensure inclusivity, the consultation venues were chosen centrally and made accessible to people of all castes, creeds, and sects in the society, ensuring that the location was considered safe by everyone in the community. Focus Group Discussions were conducted specifically to provide women and marginalized groups with an opportunity to express their opinions. |
| Working conditions | No risk identified | Occupational Health and Safety will be prioritized at every stage of the project. Employees will receive regular training to ensure their health and safety in the workplace. |

| | | |
|---|---|---|
| | | Measures will also be implemented to safeguard the health of the community near the project area. The project is committed to complying with national Occupational Health and Safety Standards [7], national labour standards[8], IFC PS2[9] and International Labour Organization Labor Standards, following all guidelines and standards. |
| Safety of women and girls | No risk identified | The project promotes safe and secure environments for all stakeholders including women and girls.<br>The proposed project is a renewable energy grid connected power project and therefore it’s not gender sensitive. To further ensure this project does not adversely impact the safety of women and girls, during the implementation of the project, all personnel involved in the project are regularly propagate in gender sensitivity, including awareness of gender discrimination and gender-based violence, and how to prevent and respond to these issues. The project has established effective monitoring and reporting mechanisms so that women can report any safety issues or acts of gender discrimination, ensuring that their voices are adequately addressed and protected. |
| Safety of minority and marginalized groups, including children | No risk identified | The project activity does not impact minority and marginalized groups including children. The project strictly complies with national legislation that explicitly forbids both forced labour and child labour. |
| Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical | No risk identified | The project is a hydro power project, hence will not result in release of pollutants to the environment.<br>The project will have a temporary impact on the environment only during the construction phase due to soil disruption, particulate dust |

[7] https://natlex.ilo.org/dyn/natlex2/r/natlex/fe/details?p3_isn=73666

[8] https://natlex.ilo.org/dyn/natlex2/r/natlex/fe/details?p3_isn=96369

[9] https://www.ifc.org/content/dam/ifc/doc/2010/2012-ifc-performance-standard-2-en.pdf

| | | |
|---|---|---|
| pesticides and fertilizers) | | generation and noise associated with vehicles, respective measures have been taken to minimize the impact. |

## 2.3 Respect for Human Rights and Equity

### 2.3.1 Labor and Work

| | Risks identified[10] | Mitigation or preventative measure(s) taken |
|---|---|---|
| Discrimination | No risk identified | The project does not involve or engage in any form of discrimination based on gender, race, religion, sexual orientation, or any other grounds. This commitment underscores the project's dedication to promoting equality and inclusivity by refraining from practices that may unfairly disadvantage any individual or group. |
| Sexual harassment | No risk identified | The project emphasizes its commitment to creating a safe and respectful work environment free from sexual harassment. Since its inception, there have been no complaints of sexual harassment. Moreover, the project has proactively established a monitoring mechanism specifically designed to address complaints about sexual harassment. This mechanism ensures that any instances of sexual harassment are promptly investigated and addressed, thereby preventing their occurrence in the future and reinforcing the project's dedication to maintaining a harassment-free workplace. |
| Equal pay for equal work | No risk identified | The project implements equal pay for equal work to ensure that workers' remuneration is commensurate with the amount of work they have done and their performance. We regularly evaluate each position to determine the nature of the work, the difficulty of the work, the amount of work, and the skills required for |

[10] The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

| | | |
|---|---|---|
| | | different positions, etc. We formulate a corresponding remuneration system based on the results of the job evaluation to ensure that laborers engaged in the same work or similar work receive the same labor compensation. |
| Gender equity in labor and work | No risk identified | The project has male and female employees of different genders, and there are female employees in management positions. There is no significant gender difference between female employees and male employees in terms of salary, promotion opportunities and welfare benefits, and employees of different genders have equal opportunities to obtain training and development resources. All employees feel that the project is treated equally in terms of gender and pay. |
| Forced labor | No risk identified | Nam Kiao Hydropower Sole Co., Ltd. adheres to international labor standards, particularly those set by the International Labor Organization (ILO), to ensure that its recruitment and employment policies do not involve any form of forced labor. Nam Kiao Hydropower Sole Co., Ltd. conducts comprehensive background checks on all employees and complies with local labor laws and regulations to prevent any exploitation or abuse. Additionally, the project has implemented a supervision mechanism, including regular on-site inspections and audits, to identify and immediately address any potential violations of labor rights, including forced labor. |
| Child labor | No risk identified | In this project, child labor is unacceptable and strictly prohibited. The project adheres to the relevant standards of the International Labour Organization as well as the local laws and regulations that prohibit the employment of minors. At the same time, the project has established a supervision mechanism and conducts regular on-site inspections and audits to ensure that there is no child labor in our operations. Nam Kiao Hydropower Sole Co., Ltd. |

| | | |
|---|---|---|
| | | also provides training for employees and partners to enhance their awareness of the harm caused by child labor and encourages them to report any suspicious situations. |
| **Human trafficking** | No risk identified | The project firmly opposes human trafficking. The project has a compliant recruitment and employment policy that complies with the standards of the International Labour Organization and does not involve any form of human trafficking. At the same time, it conducts strict background checks on all employees and partners to identify and prevent any possible acts related to human trafficking. The project also implements supervision mechanisms, including regular on-site inspections and audits, to ensure that there is no human trafficking in the business and that the rights of all individuals are respected. Nam Kiao Hydropower Sole Co., Ltd. also provides training for employees and partners to enhance their understanding of human trafficking and educate them on how to avoid and report possible violations. |

### 2.3.2 Human Rights

| Risks identified | Mitigation or preventative measure(s) taken |
|---|---|
| No risk identified. | The project adheres to human rights principles, ensuring the protection of all stakeholders. The project promotes gender equality, prevents discrimination, and prohibits child and forced labor. It follows national labor laws and international standards, including the Universal Declaration of Human Rights and ILO conventions. Equal pay for equal work is enforced, and all workers are provided with safe and respectful working conditions. The project is committed to fostering an inclusive environment for women, marginalized groups, and vulnerable populations, ensuring their |

| | |
|---|---|
| | rights are protected throughout the project lifecycle. |

### 2.3.3 Indigenous Peoples and Cultural Heritage

| Risks identified | Mitigation(s) or preventative measure taken |
|---|---|
| No risk identified. | This project utilizes hydropower. During the implementation and operation phases of the project, the project activities created employment opportunities for skilled workers and unskilled workers.<br>The project owner has secured legal land use rights; to address historical customary usage for natural resource collection, fair compensation was provided to ensure no adverse impact on stakeholders. The proponent remains committed to upholding indigenous rights through transparency, ensuring positive impacts on the ecosystem and local livelihoods. |

### 2.3.4 Property Rights

| Risks identified | Mitigation or preventative measure(s) taken |
|---|---|
| No risk identified. | This project utilizes hydropower. During the implementation and operation phases of the project, the project activities created employment opportunities for skilled workers and unskilled workers.<br>The project owner has formally obtained the land use rights for the project area, hence no property right of any stakeholders is impacted due to project activity. |

### 2.3.5 Benefit Sharing

| | |
|---|---|
| Process used to design the benefit sharing plan | Not applicable as the project owner has formally obtained the land use rights for the project area. There, is no explicit benefit sharing arrangement with the local stakeholders of any kind neither is any such arrangement required as per local laws. However, Nam Kiao Hydropower |

| | |
|---|---|
| | Sole Co., Ltd. plans to help nearby villagers through CSR activities in education, sports and other basic needs as per mutual agreement. |
| Summary of the benefit sharing plan | Not applicable. |
| Approval and dissemination of benefit sharing plan | Not applicable. |

## 2.4 Ecosystem Health

| | Risks identified | Mitigation or preventative measure(s) taken |
|---|---|---|
| Impacts on biodiversity and ecosystems | No risk identified | This project is of the run-of-the-river type and the construction process does not involve deep well drilling. Hence, groundwater is not expected to the impacted during construction. Even during the operation the water, the flow downstream will take the natural course of the river after the utilizing the water without any impact on the groundwater. The Monitoring Plan accounts for measurement the minimal water discharge and ground water level at the site periodically during the operation phase. Furthermore, desilting of the reservoir area will be conducted every 3-5 years to ensure optimal water storage capacity. |
| Soil degradation and soil erosion | No risk identified | There is no negative impact due to project instances on soil. Moreover, the project plans plantation activity near project sites to prevent any potential soil erosion. |
| Water consumption and stress | No risk identified | There are no competing uses for water resources in the Project locations. As the project is a run-of-the-river project, it will divert water stream for short distance and onwards releases the same to main river stream; this has no impact on water availability and does not create water stress. |

| | | |
|---|---|---|
| | | Furthermore, in line with the conservative practices of design, minimum discharge is maintained in river stream, which shall meet any other water requirements in the region for other social needs. |

### 2.4.1 Rare, Threatened, and Endangered Species

*Is the project located in or adjacent to habitats for rare, threatened, or endangered species?*

☐ Yes ☒ No

### 2.4.2 Introduction of Species

The project activity is not a AFOLU project, hence not applicable.

### 2.4.3 Ecosystem Conversion

The project activity is not a AFOLU project, hence not applicable.

# 3 APPLICATION OF METHODOLOGY

## 3.1 Title and Reference of Methodology

| Type (methodology, tool or module). | Reference ID, if applicable | Title | Version |
|---|---|---|---|
| **Methodology** | AMS-I.D. | Grid connected renewable electricity generation | 18.0 |
| **CDM-Tool** | Tool 7 | Tool to calculate the emission factor for an electricity system | 7.0 |
| **CDM-Tool** | Tool 21 | Demonstration of additionality of small-scale project activities | 13.1 |
| **CDM-Tool** | Tool 27 | Investment Analysis | 14.0 |

## 3.2 Applicability of Methodology

The project activity involves generation of electricity by the means of renewable energy, hydro energy having installed capacity 11 MW, which is below the 15MW threshold . Hence, the project is a type I project and the project activity falls under the category of small-scale projects. The methodology chosen for the project activity and its applicability to the project activity is discussed below.

| Methodology ID | Applicability condition | Justification of compliance |
|---|---|---|
| AMS-I.D. | This methodology comprises renewable energy generation units, such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass:<br>(a) Supplying electricity to a national or a regional grid; or<br>(b) Supplying electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling. | The Project activity involves electricity generation using hydro power plant and supply of electricity to the national grid of Vietnam.<br>Hence, criterion met. |
| AMS-I.D. | Illustration of respective situations under which each of the methodology (i.e. “AMS-I.D.: Grid connected renewable electricity generation”, “AMS-I.F.: Renewable electricity generation for captive use and mini-grid” and “AMS-I.A.: Electricity generation by the user) applies is included in the appendix. | According to the point 1 of the Table 2 in the methodology – “Project supplies electricity to a national/ regional grid” is applicable under AMS I.D.<br>As the project activity supplies the electricity to the national grid of Vietnam, which is a national grid, the methodology AMS-I.D. is applicable. |
| AMS-I.D. | This methodology is applicable to project activities that:<br>(a) Install a Greenfield plant;<br>(b) Involve a capacity addition in (an) existing plant(s);<br>(c) Involve a retrofit of (an) existing plant(s);<br>(d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or<br>(e) Involve a replacement of (an) existing plant(s). | The Project activity involves the installation of new hydro power plant at a site where there was no renewable energy power plant operating prior to the implementation of the project activity. Thus, Project activity is a Greenfield plant and satisfies this applicability condition (a). |

| | | |
|---|---|---|
| AMS-I.D. | Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology:<br><br>(a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir;<br><br>(b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4 $W/m^2$;<br><br>(c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 $W/m^2$. | The proposed project is a greenfield project, which results in a new reservoir and the power density as per definitions given in the project emissions section, is 4298.55 $W/m^2$, which is greater than 4 $W/m^2$.<br><br>Hence, Nam Kiao hydro power plant with a new reservoir that satisfy the (c) condition is eligible to apply this methodology. |
| AMS-I.D. | If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW. | The project activity is only 11 MW hydro energy based renewable electricity generation project.<br><br>It does not include any non-renewable unit and co-firing system. |
| AMS-I.D. | Combined heat and power (co-generation) systems are not eligible under this category. | This is not relevant to the project activity as the project involves only hydro power generating units. |
| AMS-I.D. | In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be | The project activity is a greenfield project.<br><br>Hence this criterion is not applicable. |

| | | |
|---|---|---|
| | physically distinct1 [11] from the existing units. | |
| AMS-I.D. | In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW. | The project activity is not the retrofitting or replacement of an existing facility for renewable energy generation.<br><br>Hence this criterion is not applicable. |
| AMS-I.D. | In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as "AMS-I.C.: Thermal energy production with or without electricity" shall be explored. | The proposed project activity is a hydroelectric power project. Hence, the criterion not applicable. |
| AMS-I.D. | In case biomass is sourced from dedicated plantations, the applicability criteria in the tool "Project emissions from cultivation of biomass" shall apply. | The proposed project activity is a hydroelectric power project. Hence, the criterion not applicable. |
| Tool 7 | This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been | This condition is applicable. OM, BM and CM are estimated using the tool under section 4.1 for calculating baseline emissions. |

[11] Physically distinct units are those that are capable of generating electricity without the operation of existing units, and that do not directly affect the mechanical, thermal, or electrical characteristics of the existing facility. For example, the addition of a steam turbine to an existing combustion turbine to create a combined cycle unit would not be considered "physically distinct".

| | | |
|---|---|---|
| | provided by the grid (e.g. demand-side energy efficiency projects). | Therefore, the tool is applicable. |
| Tool 7 | Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in "Appendix 1: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity. | Since the project activity is grid connected, this condition is applicable to the project activity and the emission factor has been calculated accordingly.<br><br>Therefore, the tool is applicable. |
| Tool 7 | In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country. | The project activity is located in Lao PDR and generated electricity supplied to Vietnam grid, both are non-Annex I countries.<br><br>Therefore, this tool is applicable for the project activity. |
| Tool 7 | Under this tool, the value applied to the CO2 emission factor of biofuels is zero. | The project does not involve biofuels usage.<br><br>Therefore, this criterion is not applicable for the project activity. |

| | | |
|---|---|---|
| Tool 21 | The use of the methodological tool "Demonstration of additionality of small-scale project activities" is not mandatory for project participants when proposing new methodologies. Project participants and coordinating/managing entities may propose alternative methods to demonstrate additionality for consideration by the Executive Board. | The project proponent is not proposing any new methodology and has applied this tool for the demonstration additionality in compliance with the tool. Refer to section 3.5, for the detailed applicability of this tool and additionality assessment.<br><br>Hence this tool is applicable. |
| Tool 21 | Project participants and coordinating/managing entities may also apply " TOOL19: Demonstration of additionality of microscale project activities" as applicable. | The project is not a microscale project.<br><br>Therefore, this criterion is not applicable for the project activity. |
| Tool 27 | This methodological tool is applicable to CDM project activities and programmes of activities (PoAs) that conduct an investment analysis for the demonstration of additionality and/or the identification of the baseline scenario. | The investment analysis has been conducted in accordance with Tool 27. For details, please refer to Section 3.5.2.<br><br>Hence, the tool is applicable. |
| Tool 27 | In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail. | The investment analysis method selected by the project is consistent with the requirements of Tool 27. Please refer to Section 3.5.2 for the detailed analysis.<br><br>Hence, the tool is applicable. |

## 3.3 Project Boundary

This section will be completed in the full Project Description (PD) submitted for the 'under validation' status, as this version of the PD is for pipeline listing purposes only.

| Source | | Gas | Included? | Justification/Explanation |
|---|---|---|---|---|
| Baselin | | $CO_2$ | | |

| Source | | Gas | Included? | Justification/Explanation |
|---|---|---|---|---|
| | | $CH_4$ | | |
| | | $N_2O$ | | |
| Project | | $CO_2$ | | |
| | | $CH_4$ | | |
| | | $N_2O$ | | |

## 3.4 Baseline Scenario

The proposed project activity involves installation and operation of 11 MW greenfield hydro power project. The generated power will be supplied to Vietnam Grid, which otherwise would have been generated from grid connected power plants, which possesses a mix of generation types with fossil fuel fired power plants.

As per para 19 of AMS-I.D. (Version 18) "The baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid." In the absence of the project activity same amount electricity would have been generated from grid, in which the electricity is generated by the fossil fuel intensive power plant. Thus, baseline is in line with para 19 of AMS-I.D. (Version 18).

Para 22 of AMS-I.D. (Version 18) calculates baseline emissions as:

$$BE_y = EG_{PJ,y} \times EF_{grid,y} \quad \text{Equation (1)}$$

Where,

| | | |
|---|---|---|
| $BE_y$ | = | Baseline emissions in year y (t $CO_2$) |
| $EG_{PJ,y}$ | = | Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh) |
| $EF_{grid,y}$ | = | Combined margin $CO_2$ emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (t $CO_2$/MWh) |

Accordingly, the emission factor of the grid will be used to estimate emission reductions. As per para 23 of AMS-I.D. (Version 18), Nam Kiao Hydropower Sole Co., Ltd. has chosen option (a) and used the combined margin (CM) approach to calculate emission factor, as official data is available for operating margin (OM) and build margin (BM) values, whereas no such data exists in the public domain to support choice of option (b).

Hence,

$$EF_{grid,y} = EF_{grid,CM,y}$$

The combined margin emission factor of the grid used for the project activity is as follows:

| Parameter | Value | Description |
|---|---|---|
| $EF_{grid,CM,y}$ | 0.6592 $tCO_2$/MWh | Combined margin $CO_2$ emission factor for the project electricity system in year y |
| $EF_{grid,OM,y}$ | 0.9201 $tCO_2$/MWh | Operating margin $CO_2$ emission factor for the project electricity system in year y |
| $EF_{grid,BM,y}$ | 0.3983 $tCO_2$/MWh | Build margin $CO_2$ emission factor for the project electricity system in year y |

## 3.5 Additionality

*Demonstrate and assess the additionality of the project, in accordance with the applied methodology and any relevant tools, taking into account the following additionality methods:*

### 3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☒ Yes ☒ No

The project activity is not enforced by any law, statute, or regulatory framework in Lao PDR.

The project passes the regulatory surplus test as there are no enforced laws, regulations, court orders, or environmental mitigation agreements that mandate the implementation of this specific hydro power project. Any concession agreements or power purchase agreements signed by the project owner are voluntary commercial commitments and do not constitute a legal mandate for the purpose of additionality.

While mandatory laws and regulations exist in the sector, they regulate how the project is constructed and operated, not whether it must be implemented. These include:

Law on Electricity (amended version) No. 03/NA, dated 20/12/2011[12] ;

Decree on the Approval and Promulgation of the Policy on Sustainable Hydropower Development of the Lao PDR No. 02/GoL., dated 12/01/2015[13] ;

Environmental Protection Law (revised version), No. 29/NA, dated 18/12/2012[14];

National Policy on Environmental and Social Sustainability of the Hydropower Sector in Lao PDR[15].

None of the above-referenced regulations mandate the implementation of hydro power project.

### 3.5.2 Additionality Methods

This section will be completed in the full Project Description (PD) submitted for the 'under validation' status, as this version of the PD is for pipeline listing purposes only.

## 3.6 Methodology Deviations

There is no methodology deviation.

# 4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

## 4.1 Baseline Emissions

According to methodology AMS-I.D. (Version 18.0), baseline emissions include only $CO_2$ emissions from electricity generation in power plants that are displaced due to the project activity. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,y}$$

Where:

---

[12] https://policy.asiapacificenergy.org/sites/default/files/Electricity%20Law.pdf

[13] policy.asiapacificenergy.org/sites/default/files/Decree%20on%20the%20Approval%20and%20Promulgation%20of%20the%20Policy%20on%20Sustainable%20Hydropower%20Development%20in%20Lao%20PDR.pdf

[14] https://www.ajcsd.org/chrip_search/dt/pdf/Useful_materials/Lao_PDR/Environmental_Protection_Law_Revised_Version.pdf

[15] policy.asiapacificenergy.org/sites/default/files/National%20Policy%20on%20the%20Environmental%20and%20Social%20Sustainability%20of%20the%20Hydropower%20Sector.pdf

| | | |
|---|---|---|
| $BE_y$ | = | Baseline emissions in year y (t $CO_2$) |
| $EG_{PJ,y}$ | = | Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh) |
| $EF_{grid,y}$ | = | Combined margin $CO_2$ emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (t $CO_2$/MWh) |

**Calculation of $EG_{PJ,y}$**

Since the project activity consists of the installation of new grid-connected renewable power plant at a site where no renewable power plant was operated prior to the implementation of the project activity, the project activity is the installation of a greenfield renewable energy power plant, so that:

$$EG_{PJ,y} = EG_{PJ,facility,y}$$

Where:

| | | |
|---|---|---|
| $EG_{PJ,facility,y}$ | = | Quantity of net electricity generation supplied by the project plant to the grid in year y (MWh) |

**Calculation of $EF_{grid,y}$**

The emission factor should be calculated in a transparent and conservative manner according to the procedures prescribed in the Tool 07 "Tool to calculate the emission factor for an electricity system" (Version 07.0).

The data used for calculation are from an official source (where available) and publicly available. The calculation processes are as follows:

Step1: Identify the relevant electricity system;

Step2: Choose whether to include off-grid power plants in the project electricity system (optional);

Step3: Select a method to determine the operating margin (OM);

Step4: Calculate the operating margin emission factor according to the selected method;

Step5: Calculate the build margin (BM) emission factor;

Step6: Calculate the combined margin (CM) emissions factor.

**Step1: Identify the relevant electricity system**

This project will be connected to the national electricity grid of Viet Nam, which is operated and monopolized by the EVN. This national electricity grid is the unique transmission and distribution line, to which all power plants in Vietnam are physically connected. Hence, the national electricity grid is the project electricity system.

There are electricity imports to the Viet Nam national electricity grid from China and Laos, thus the China and Laos Power Grid is the connected electricity systems. As per Tool 07, for the purpose of determining the operating margin emission factor, the $CO_2$ emission factor for net electricity imports from connected electricity systems is 0 $tCO_2/MWh$ by default.

**Step2: Choose whether to include off-grid power plants in the project electricity system (optional)**

There are two options to calculate the operating margin and build margin emission factor.

Option1: Only grid power plants are included in the calculation;

Option 2: Both grid power plants and off-grid power plants are included in the calculation

Because only the data of grid connected power plants is available, so Option I will be chosen for calculating the grid emission factor.

**Step3: Select a method to determine the operating margin (OM)**

The calculation of the operating margin emission factor ($EF_{grid,OM,y}$) is based on one of the following methods, which are described under Step4:

(a) Simple OM; or

(b) Simple adjusted OM; or

(c) Dispatch data analysis OM; or

(d) Average OM.

The method (a) "Simple OM" was chosen for Viet Nam because the low cost/must-run resources constitute less than 50% of the total grid generation in the most recent five years (detail see in the table below):

Table 4-1: Rate of low cost/must-run resources based on generation.

| Year | 2019 | 2020 | 2021 | 2022 | 2023 | total |
|---|---|---|---|---|---|---|
| Hydropower | 54,411,106 | 59,387,446 | 69,606,845 | 78,225,090 | 70,469,947 | 333,702,417 |
| Bagasse power | 280,996 | 331,319 | 347,560 | 246,621 | 868,000 | 2,074,496 |
| Wind power | 721,189 | 946,157 | 3,243,227 | 8,902,337 | 12,481,822 | 26,414,247 |
| Solar power | 4,833,674 | 9,684,525 | 15,141,520 | 14,381,800 | 16,083,045 | 60,124,564 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Import | 3,316,000 | 3,067,000 | 1,401,463 | 1,357,845 | 4,220,000 | 15,275,308 |
| Total grid generation | 207,214,694 | 207,692,796 | 208,561,267 | 220,446,789 | 236,285,221 | / |
| Low-cost/Must-run ratio | | | | | | 40.51% |

The Ex-ante option was chosen, thus, no monitoring and recalculation of the emissions factor during the crediting period is required. The 3-year generation-weighted average (2021, 2022 and 2023) that is the most recent data available was used to calculate the OM emission factor.

**Step 4: Calculate the operating margin emission factor according to the selected method**

The simple OM emission factor is calculated as the generation-weighted average $CO_2$ emissions per unit net electricity generation ($tCO_2/MWh$) of all generating power plants serving the system, not including low-cost / must-run power plants / units.

The simple OM may be calculated by one of the following two options:

**Option A:** Based on the net electricity generation and a $CO_2$ emission factor of each power unit; Or

**Option B:** Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system.

The Option B was chosen by DNA of Viet Nam, "Viet Nam national grid emission factor report 2023" (attached with the Official Letter No. 1726/BDKH-PTCBT" published on 3/12/2024, in line with Tool 07 because the necessary data for Option B is available. Under Option B, the simple OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, not including low cost/must run power plants/units, and based on the fuel type(s) and total fuel consumption of the project electricity system, as follows:

$$EF_{grid,OMsimple,y} = \sum \left( FC_{i,y} \times NCV_{i,y} \times EF_{CO2,i,y} \right) / EG_y$$

Where,

| | | |
|---|---|---|
| $EF_{grid,OMsimple,y}$ | = | Simple operating margin $CO_2$ emission factor in year y ($tCO_2/MWh$) |
| $FC_{i,y}$ | = | Amount of fuel type i consumed in the project electricity system in year y(mass or volume unit) |
| $NCV_{i,y}$ | = | Net calorific value (energy content) of fuel type i in year y (GJ/mass or volume unit) |
| $EF_{CO2,i,y}$ | = | $CO_2$ emission factor of fossil fuel type i in year y ($tCO_2/GJ$) |
| $EG_y$ | = | Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must-run power plants/units, in year y (MWh) |

| | | |
|---|---|---|
| i | = | All fuel types combusted in power sources in the project electricity system in year y |
| y | = | The relevant year as per the data vintage chosen in Step 3 |

On the basis of the data available, the three-year average operating margin emission factor is calculated by the DNA as a full-generation-weighted average of the emission factors:

$EF_{grid,OM,y}$ = 0.9201 $tCO_2$/MWh

Step 5: Calculate the build margin emission factor ($EF_{BM,y}$)

In terms of vintage of data, project participants can choose between one of the following two options:

Option1: For the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of CDM-PDD submission to the DOE for validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the DOE. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period;

Option2: For the first crediting period, the build margin emission factor shall be updated annually, ex post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex ante, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

The Option 1 was chosen by DNA of Viet Nam, "Viet Nam national grid emission factor report 2023" (attached with the Official Letter No. 1726/BDKH-PTCBT" published on 3/12/2024. The BM emission factor is calculated ex ante based on the most recent information available on units already built for sample group m at the time of this project description submission.

The sample group of power units m used to calculate the build margin should be determined as per the following procedure, consistent with the data vintage selected above:

(a)Identify the set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently ($SET_{5\text{-}units}$) and determine their annual electricity generation ($AEG_{SET\text{-}5\text{-}units}$, in MWh);

The set of five most recently built power units ($SET_{5\text{-}units}$) is indicated in Appendix X. These five power units generated 220,141.6 MWh of electricity ($AGE_{SET\text{-}5\text{-}units}$) in 2023.

(b) Determine the annual electricity generation of the proposed project electricity system, excluding power units registered as CDM project activities ($AEG_{total}$, in MWh). Identify the set of power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently and that comprise 20% of $AEG_{total}$ (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) (SET$\geqslant$20%) and determine their annual electricity generation ($AEG_{SET\geqslant 20\%}$, in MWh);

The total output of Viet Nam electricity grid ($AEG_{total}$) in 2023 is 47,402,236.48 MWh. The most recent-built power plants (SET$\geqslant$20 per cent)selected by DNA of Viet Nam accounted for 20.06% of the system electricity generation ($AEG_{total}$) and generated 236,285,220.98 MWh of electricity ($AEG_{SET-\geqslant 20\%}$) in 2023.

(c) From $SET_{5\text{-units}}$ and $SET_{\geqslant 20\ percent}$ select the set of power units that comprises the larger annual electricity generation ($SET_{sample}$).

$SET_{\geqslant 20\ percent}$ (236,285,220.98 MWh) is larger than $SET_{5\text{-units}}$ (220,141.6 MWh); and none of the power units in $SET_{\geqslant 20\ percent}$ started to supply electricity to the grid more than 10 years ago, then $SET_{\geqslant 20\ percent}$ is chosen as $SET_{sample}$ and use $SET_{sample}$ to calculate the build margin.

The build margin emissions factor is the generation-weighted average emission factor ($tCO_2$/MWh) of all power units m during the most recent year y for which power generation data is available. It is calculated as follows:

$$EF_{grid,BM,y} = \frac{\sum_m EG_{m,y} \times EF_{EL,m,y}}{\sum_m EG_{m,y}}$$

Where,

| | | |
|---|---|---|
| $EF_{grid,BM,y}$ | = | Build margin $CO_2$ emission factor in year y (t CO2/MWh) |
| $EG_{m,y}$ | = | Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh) |
| $EF_{EL,m,y}$ | = | $CO_2$ emission factor of power unit m in year y($tCO_2$/MWh) |
| m | = | Power units included in the build margin |
| y | = | Most recent historical year for which power generation data is available |

$EF_{grid,BM,y}$ = 0.3983 $tCO_2$/MWh

Step 6: Calculate the combined margin emissions factor

According to Tool 07, the calculation of the combined margin (CM) emission factor ($EF_{grid,CM,y}$) is based on one of the following methods:

(a) Weighted average CM; or

(b) Simplified CM

The weighted average CM method (option a) is the preferred option and used to calculate the CM emission factor of the national gird.

The combined margin emissions factor is calculated as follows:

$$EF_{grid,CM,y} = EF_{\mathrm{grid},OM,y} \times \omega_{OM} + EF_{grid,CM,y} \times \omega_{BM}$$

Where,

| | | |
|---|---|---|
| $EF_{grid,BM,y}$ | = | Build margin $CO_2$ emission factor in year y (t $CO_2$/MWh) |
| $EF_{grid,OM,y}$ | = | Operating margin $CO_2$ emission factor in year y (t $CO_2$/MWh) |
| $\omega_{OM}$ | = | Weighting of operating margin emissions factor (per cent) |
| $\omega_{BM}$ | = | Weighting of build margin emissions factor (per cent) |

Except for wind and solar power generation project activities, the operating margin and build margin for all other projects are 50% and 50%. As the project activity is hydro power project the above weight is applied to calculated combined margin emission factor.

Therefore,the baseline emission factor $EF_y$ is calculated as the weighted average of the Operating Margin emission factor ($EF_{OM,y}$) and the Build Margin emission factor ($EF_{BM,y}$) as below:

$EF_{grid,CM,y}$ = 0.9201 x 0.5 + 0.3983 x 0.5=0.6592 $tCO_2$/MWh

Hence the baseline emission is calculated as below:

$BE_y$= 36,530 × 0.6592 = 24,080 $tCO_2$

## 4.2 Project Emissions

The project activity is a hydro power project activity and as per the para 39 of the applied methodology the Emissions from water reservoirs of hydro power plants shall be determined considered following the procedure described in the most recent version of “ACM0002: Grid connected electricity generation from renewable sources”. The project emissions are to be calculated as follows:

$$PE_y = PE_{FF,y} + PE_{HP,y} + PE_{PSP,y}$$

Where,

| | | |
|---|---|---|
| $PE_y$ | = | Project emissions in year y (t $CO_2$e/yr) |
| $PE_{FF,y}$ | = | Project emissions from fossil fuel consumption in year y (t $CO_2$/yr) |
| $PE_{HP,y}$ | = | Project emissions from water reservoirs of hydro power plants amd pumped storage plants in year y (t $CO_2$e/yr) |

| | | |
|---|---|---|
| $PE_{PSP,y}$ | = | Project emissions from utilizing electricity from the grid for pumping operation of PSP in excess to the production of the renewable power plant operating in coordination with the PSP (t $CO_2e$/yr) |

**Emissions from fossil fuel combustion($PE_{FF,y}$)**

As per the para 39 of the applied methodology, all renewable energy power generation project activities, emissions due to the use of fossil fuels for the backup generator can be neglected.

**Emissions from water reservoirs of hydro power plants and pumped storage projects ($PE_{HP,y}$)**

The power density (PD) of the project activity is calculated as follows:

$$PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$$

Where,

| | | |
|---|---|---|
| PD | = | Power density of the project activity (W/$m^2$) |
| $Cap_{PJ}$ | = | Installed capacity of the hydro power plant or PSP after the implementation of the project activity (W) |
| $Cap_{BL}$ | = | Installed capacity of the hydro power plant or PSP before the implementation of the project activity (W). For new hydro power plants or PSP, this value is zero. |
| $A_{PJ}$ | = | Area of the single or multiple reservoirs measured in the surface of the water, after the implementation of the project activity, when the reservoir is full ($m^2$) |
| $A_{BL}$ | = | Area of the single or multiple reservoirs measured in the surface of the water, before the implementation of the project activity, when the reservoir is full ($m^2$). For new reservoirs, this value is zero. |

PD = (11,000,000-0)/(2559-0) = 4298.55W/$m^2$

As per ACM0002(Version 22.0), If the power density of the project activity is greater than 10 W/$m^2$， $PE_{HP,y}$ =0 $tCO_2$.

**Emissions from utilizing grid electricity by pumped hydro plants ($PE_{PSP,y}$)**

Since the project does not consume electricity from the power grid, $PE_{PSP,y}$=0 $tCO_2$.

In conclusion, the project emission of the project activity is 0 $tCO_2$.

## 4.3 Leakage Emissions

As per applied approved methodology no leakage emission relevant to project activity. $LE_y$ =0 $tCO_2$.

## 4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

The ex-ante emission reductions ($ER_y$) for the project activity are calculated as follows:

$$ER_y = BE_y - PE_y - LE_y$$

Where,

| | | |
|---|---|---|
| $ER_y$ | = | Emission reductions in year y (t $CO_2$) |
| $BE_y$ | = | Baseline Emissions in year y (t $CO_2$) |
| $PE_y$ | = | Project emissions in year y (t $CO_2$) |
| $LE_y$ | = | Leakage emissions in year y (t $CO_2$) |

**Emissions reductions are calculated as:**

$ER_y$=24,080 $tCO_2e$/yr - 0 $tCO_2e$/yr - 0 $tCO_2e$/yr = 24,080 $tCO_2e$/year

The emission reductions can be obtained as below:

Table 4-2 The estimation of $ER_y$

| Vintage period | Estimated baseline emissions ($tCO_2e$) | Estimated project emissions ($tCO_2e$) | Estimated leakage emissions ($tCO_2e$) | Estimated reduction VCUs ($tCO_2e$) | Estimated removal VCUs ($tCO_2e$) | Estimated total VCUs ($tCO_2e$) |
|---|---|---|---|---|---|---|
| 31-Mar-2026 to 31-Dec-2026 | 18,208 | 0 | 0 | 18,208 | 0 | 18,208 |
| 01-Jan-2027 to 31-Dec-2027 | 24,080 | 0 | 0 | 24,080 | 0 | 24,080 |
| 01-Jan-2028 to 31-Dec-2028 | 24,080 | 0 | 0 | 24,080 | 0 | 24,080 |
| 01-Jan-2029 to 31-Dec-2029 | 24,080 | 0 | 0 | 24,080 | 0 | 24,080 |
| 01-Jan-2030 to 31-Dec-2030 | 24,080 | 0 | 0 | 24,080 | 0 | 24,080 |
| 01-Jan-2031 to 31-Dec-2031 | 24,080 | 0 | 0 | 24,080 | 0 | 24,080 |
| 01-Jan-2032 to 31-Dec-2032 | 24,080 | 0 | 0 | 24,080 | 0 | 24,080 |

| | | | | | | |
|---|---|---|---|---|---|---|
| 01-Jan-2033 to 31-Dec-2033 | 24,080 | 0 | 0 | 24,080 | 0 | 24,080 |
| 01-Jan-2034 to 31-Dec-2034 | 24,080 | 0 | 0 | 24,080 | 0 | 24,080 |
| 01-Jan-2035 to 31-Dec-2035 | 24,080 | 0 | 0 | 24,080 | 0 | 24,080 |
| 01-Jan-2036 to 30-Mar-2036 | 5,871 | 0 | 0 | 5,871 | 0 | 5,871 |
| **Total** | 240,800 | 0 | 0 | 240,799 | 0 | 240,799 |

# 5 MONITORING

## 5.1 Data and Parameters Available at Validation

This section will be completed in the full Project Description (PD) submitted for the 'under validation' status, as this version of the PD is for pipeline listing purposes only.

| | |
|---|---|
| Data / Parameter | |
| Data unit | |
| Description | |
| Source of data | |
| Value applied | |
| Justification of choice of data or description of measurement methods and procedures applied | |
| Purpose of data | |
| Comments | |

## 5.2 Data and Parameters Monitored

This section will be completed in the full Project Description (PD) submitted for the 'under validation' status, as this version of the PD is for pipeline listing purposes only.

| | |
|---|---|
| Data / Parameter | |
| Data unit | |
| Description | |
| Source of data | |
| Description of measurement methods and procedures to be applied | |
| Frequency of monitoring/recording | |
| Value applied | |
| Monitoring equipment | |
| QA/QC procedures to be applied | |
| Purpose of data | |
| Calculation method | |
| Comments | |

## 5.3 Monitoring Plan

This section will be completed in the full Project Description (PD) submitted for the 'under validation' status, as this version of the PD is for pipeline listing purposes only.

# APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

*Use the table below to describe the commercially sensitive information included in the project description to be excluded in the public version.*

| Section | Information | Justification |
|---|---|---|
| | | |
| | | |
| | | |
| | | |

# APPENDIX X: <TITLE OF APPENDIX>

*Use appendices for supporting information. Delete this appendix (title and instructions) where no appendix is required.*